```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Using the image already available in Colab
img = cv2.imread('image1.jpg', 0)

if img is not None:
    # Calculate Sobel gradients
    sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
    sobely = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

    # Combine gradients and convert to uint8
    abs_sobelx = cv2.convertScaleAbs(sobelx)
    abs_sobely = cv2.convertScaleAbs(sobely)
    edges = cv2.addWeighted(abs_sobelx, 0.5, abs_sobely, 0.5, 0)

    # Save and display
    cv2.imwrite('Edge_detection.jpg', edges)
    print("Combined Sobel Edge Detection:")
    cv2_imshow(edges)
else:
    print("Error: 'image1.jpg' not found. Please ensure the file is uploade
```

```
Combined Sobel Edge Detection:
```